Human-like self-embedding difficulty without human-like memory constraints

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Self-embedded (SE) sentences like (1) are hard to process. Since ref. [1], this has been widely viewed as being due to performance limits on an otherwise unboundedly recursive competence grammar: too much recursion exceeds the working memory (WM) system's capacity to keep track of who did what to whom. An alternative to this ***memory account*** is a ***learning account*** based on connectionist theories that reject a distinction between performance and competence. The learning account instead derives SE difficulty from statistical experience with language: SE sentences are hard (irrespective of the processor's capacity) because they are infrequent and thus improbable [2-5]. The ideal experiment to distinguish these accounts would be to remove WM constraints on an otherwise competent language user: the memory account predicts that SE difficulty would vanish [1], whereas the learning account predicts that SE difficulty would remain [2]. This experiment is impossible, so the best testbed is computational simulation, which to-date has largely used small recurrent neural networks trained on toy datasets [2-5]. Such models have limitations for this question: they likely retain severe memory constraints [6] and do not plausibly approximate real-world human language abilities. Here, we revisited this question using Transformer [7] language models (TLMs), which overcame both these limitations: TLMs have superhuman WM and recursion capacity both in theory [8] and in simulation [9, 10], and they approximate human language so convincingly that users reliably mistake them for human [11]. Indeed, because of this, we launched this study expecting TLMs to exhibit a capacity for deep recursion [12]. Nonetheless, we found that even frontier (state-of-the-art instruction-tuned) TLMs show strikingly human-like SE difficulty, supporting the learning account.

Our study contained three parts. In ***Part 1***, we tested frontier TLMs in an offline (chain-of-thought enabled) SE grammaticality judgment task where the output was binary True/False given a sentence. Models were highly accurate at scoring both human-generated SE sentences and even very deeply recursive synthetic SE sentences – ungrammatical sentences in both sets either missed verb phrases or were semantically incoherent. Figure 1 exemplifies the performance of Claude Sonnet 4 which we selected as our auto scorer for subsequent studies owing to consistently high accuracy across recursion depths. In ***Part 2*****,** humans (N=371) and pre-trained (base) TLMs completed 24 shallow (depth 0-2) preambles (e.g., *The dollar that the biochemist that*…, Table 1) and both failed sharply at layer 2 (Fig 2). In ***Part 3*****,** (to mitigate the possibility that results in Part 2 were driven by poor instruction following in base models) humans (N=80) and frontier instruction-tuned TLMs (e.g., GPT-5, GPT-OSS, Gemma 3) with CoT disabled/minimized completed preambles with recursion depth from zero to ten (e.g., *The traveler that the scorpion that the bush that the rabbit that the snake*…, depth=4). Human (Fig 3) and model (Fig 4) accuracy dropped sharply and similarly with depth. Varying prompts and temperatures slightly modulate TLMs' performance but do not shift the overall trend. TLMs' internal probabilistic representations of verb continuations are being investigated as follow-up.

The failed prediction of the memory account in our study plausibly has other explanations: TLMs and humans may show similar patterns by different means, and SEs may be rare because WM constraints prevent humans from producing them. But our study demonstrates that the learning account is sufficient and therefore more parsimonious: an independently motivated statistical learning objective produces human-like SE difficulty even without human-like working memory constraints.

(1) The apartment that the maid who the service sent over was cleaning was well decorated.

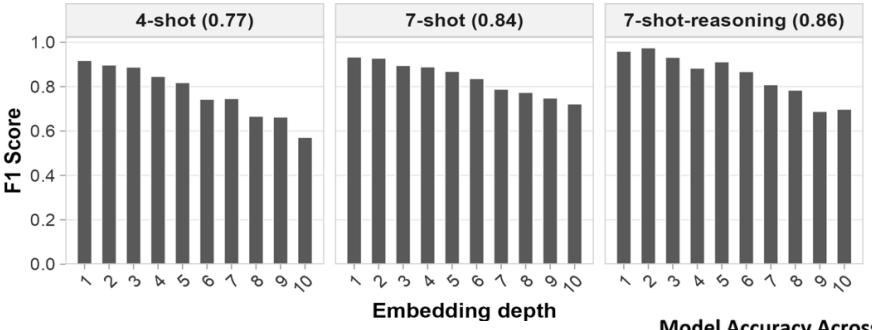

Figure 1: Annotation accuracy from Claude Sonnet 4 on synthetic data (2000 sentences with balanced True/False labels). Short titles for each figure are prompt features and the aggregate F1 score.

one-subRC (sub_ext_RC)	The biochemist who
one-objRC (obj_ext_RC)	The farmer who the
two-inanimate (RC_RC_inan)	The map which the weatherman who
two-animate (RC_RC_an)	The waiter who the customer who

Table 1: Preamble examples for shallow recursion generation in Part 2.

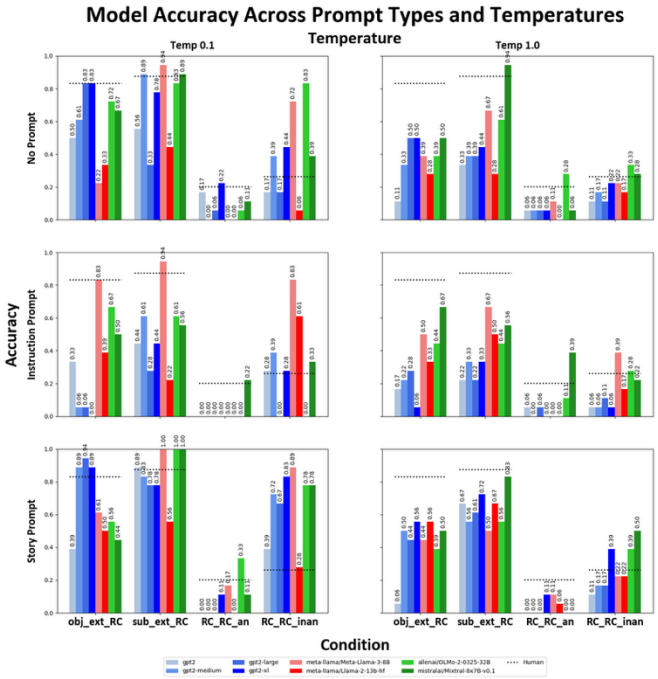

Figure 2: Generation accuracy for Part 1 from pre-trained LMs by model type, preamble condition, model temperature, and prompt type (black dotted lines represent human performance). Max new tokens = 64 [plenty for depth 2] top_p = 0.95.

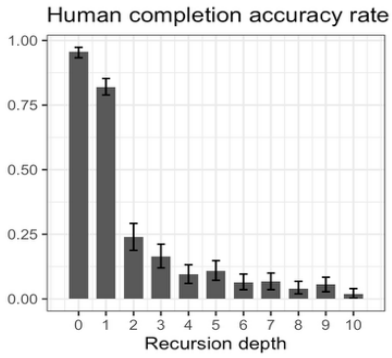

Figure 3: Generation accuracy from humans in Part 3.

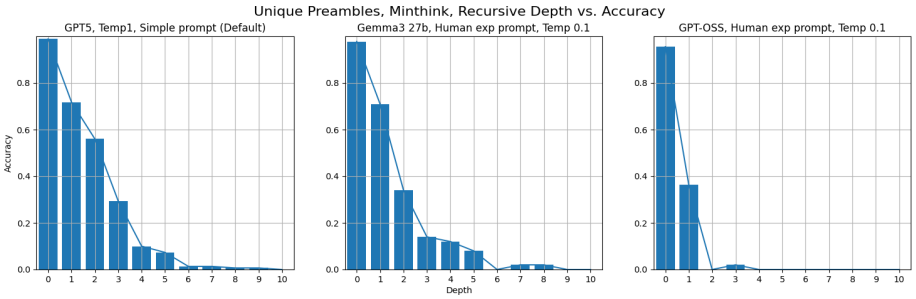

Figure 4: Generation accuracy with recursion depth for various (minimal thinking) TLMs: GPT-5 (frontier), Gemma 3 27b (open-source), and GPT-OSS 20b (open-source). For Part 3, varied temperature and prompt design (human experiment prompt [the same prompt used with human participants, utilizing multi-shot recursion completion examples] vs. simple prompt [a simple imperative sentence]).

References: [1] Miller & Chomsky 1963; [2] Christiansen & Chater 1999; [3] MacDonald & Christiansen 2002; [4] Wells et al. 2008; [5] MacDonald & Christiansen 2009; [6] Caplan & Waters 2002; [7] Vaswani et al 2017; [8] Yao et al. 2021, ACL; [9] Zhang et al 2024, TMLR; [10] Murty et al. 2023, EMNLP; [11] Jones & Bergen 2025; [12] Lampinen 2024